

PEMS03

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	14.42 ± 0.03	13.70 ± 0.02	15.14 ± 0.04	14.22 ± 0.03	10.61 ± 0.02	11.09 ± 0.03	15.25 ± 0.04	12.76 ± 0.02	11.22 ± 0.03	13.88 ± 0.04	18.42 ± 0.05	14.18 ± 0.03	13.10 ± 0.02	12.89 ± 0.01	14.28 ± 0.05	15.46 ± 0.04	13.16 ± 0.01	14.17 ± 0.04	12.67	12.53	13.20	12.13 ± 0.04	14.50	16.03 ± 0.06	13.66	12.16 ± 0.01	10.81 ± 0.02	10.78 ± 0.01	11.46 ± 0.03
	6	14.67 ± 0.04	14.89 ± 0.05	15.63 ± 0.06	15.08 ± 0.03	11.21 ± 0.01	14.01 ± 0.01	17.32 ± 0.01	13.68 ± 0.01	14.17 ± 0.01	14.76 ± 0.01	14.28 ± 0.01	15.64 ± 0.01	14.00 ± 0.01	13.82 ± 0.01	14.89 ± 0.01	17.86 ± 0.01	14.05 ± 0.01	15.45 ± 0.01	14.85	14.74	15.55	13.87 ± 0.04	17.22	18.39 ± 0.05	15.87	13.70 ± 0.02	12.22 ± 0.01	11.29 ± 0.01	12.43 ± 0.05
	12	16.27 ± 0.07	17.57 ± 0.07	17.27 ± 0.07	17.24 ± 0.03	12.66 ± 0.01	15.92 ± 0.01	19.41 ± 0.01	15.46 ± 0.01	16.10 ± 0.01	15.57 ± 0.01	16.24 ± 0.01	18.65 ± 0.01	15.86 ± 0.01	15.74 ± 0.01	16.71 ± 0.01	21.02 ± 0.01	15.95 ± 0.01	18.34 ± 0.01	20.14	19.88	20.21	17.38 ± 0.04	22.10	22.80 ± 0.05	20.35	16.86 ± 0.02	12.98 ± 0.01	12.14 ± 0.01	14.24 ± 0.08
	Avg	14.99 ± 0.05	15.15 ± 0.05	15.88 ± 0.05	15.33 ± 0.03	11.37 ± 0.01	14.16 ± 0.01	17.47 ± 0.01	13.80 ± 0.01	14.32 ± 0.01	14.90 ± 0.01	14.47 ± 0.01	15.91 ± 0.01	14.15 ± 0.01	13.98 ± 0.01	15.14 ± 0.01	18.20 ± 0.01	14.21 ± 0.01	15.73 ± 0.01	15.45	15.27	15.91	14.16 ± 0.04	17.59	18.68 ± 0.05	16.19	13.95 ± 0.02	12.26 ± 0.01	11.31 ± 0.01	12.54 ± 0.05
RMSE	3	23.64 ± 0.07	22.30 ± 0.06	23.88 ± 0.08	23.01 ± 0.03	9.61 ± 0.02	11.27 ± 0.03	25.71 ± 0.04	20.83 ± 0.02	20.39 ± 0.03	22.19 ± 0.04	20.71 ± 0.03	22.90 ± 0.04	21.23 ± 0.02	20.98 ± 0.01	25.67 ± 0.05	25.72 ± 0.04	21.37 ± 0.01	23.03 ± 0.04	21.63	21.37	22.46	20.93 ± 0.04	25.10	27.89 ± 1.00	23.21	20.77 ± 0.01	19.61 ± 0.02	18.06 ± 0.01	19.17 ± 0.07
	6	24.33 ± 0.08	24.71 ± 0.09	29.72 ± 0.09	24.70 ± 0.04	11.30 ± 0.02	16.20 ± 0.02	28.00 ± 0.02	22.70 ± 0.02	23.35 ± 0.02	24.08 ± 0.02	23.56 ± 0.02	25.90 ± 0.02	23.13 ± 0.02	22.93 ± 0.02	26.68 ± 0.02	27.99 ± 0.02	23.14 ± 0.02	25.74 ± 0.02	26.10	25.87	27.14	24.74 ± 0.04	30.33	32.52 ± 0.08	27.41	23.96 ± 0.02	20.52 ± 0.01	19.14 ± 0.01	21.09 ± 0.11
	12	27.56 ± 0.12	29.82 ± 0.12	32.46 ± 0.12	28.79 ± 0.04	14.99 ± 0.02	26.94 ± 0.02	32.37 ± 0.02	26.14 ± 0.02	27.11 ± 0.02	27.75 ± 0.02	27.50 ± 0.02	31.87 ± 0.02	26.81 ± 0.02	26.70 ± 0.02	29.78 ± 0.02	33.28 ± 0.02	26.74 ± 0.02	31.45 ± 0.02	37.11	36.36	36.42	32.31 ± 0.04	39.29	40.97 ± 0.09	35.98	30.56 ± 0.02	22.05 ± 0.01	20.82 ± 0.01	24.47 ± 0.16
	Avg	24.90 ± 0.09	25.15 ± 0.09	30.12 ± 0.09	25.15 ± 0.04	11.64 ± 0.02	18.46 ± 0.02	28.29 ± 0.02	22.89 ± 0.02	23.60 ± 0.02	24.32 ± 0.02	23.89 ± 0.02	26.40 ± 0.02	23.37 ± 0.02	23.18 ± 0.02	27.10 ± 0.02	28.58 ± 0.02	23.42 ± 0.02	26.23 ± 0.02	27.38	26.98	27.86	25.36 ± 0.04	30.92	32.99 ± 0.08	28.00	24.51 ± 0.02	20.58 ± 0.01	19.16 ± 0.01	21.25 ± 0.10
MAPE (%)	3	28.08 ± 0.09	24.72 ± 0.08	28.13 ± 0.09	26.65 ± 0.05	10.48 ± 0.02	15.11 ± 0.03	28.43 ± 0.04	24.05 ± 0.02	25.49 ± 0.03	26.87 ± 0.04	26.48 ± 0.03	27.89 ± 0.05	25.03 ± 0.02	24.96 ± 0.01	26.31 ± 0.04	29.62 ± 0.05	25.56 ± 0.02	26.28 ± 0.05	22.73	22.59	23.53	21.74 ± 0.04	23.11	23.82 ± 0.03	23.23	22.00 ± 0.02	22.52 ± 0.01	21.22 ± 0.01	23.52 ± 0.29
	6	28.30 ± 0.10	26.15 ± 0.08	28.46 ± 0.09	27.65 ± 0.04	10.81 ± 0.02	16.03 ± 0.02	29.35 ± 0.02	24.75 ± 0.02	27.29 ± 0.02	27.58 ± 0.02	27.28 ± 0.02	29.56 ± 0.02	26.82 ± 0.02	25.89 ± 0.02	26.15 ± 0.02	32.24 ± 0.02	26.53 ± 0.02	27.99 ± 0.05	25.08	25.06	26.19	23.46 ± 0.04	25.83	26.96 ± 0.03	25.68	23.83 ± 0.02	22.95 ± 0.01	21.71 ± 0.01	24.44 ± 0.26
	12	30.21 ± 0.13	29.78 ± 0.09	30.89 ± 0.09	30.44 ± 0.04	14.61 ± 0.02	23.34 ± 0.02	34.64 ± 0.02	26.62 ± 0.02	29.70 ± 0.02	29.44 ± 0.02	29.57 ± 0.02	34.17 ± 0.02	28.11 ± 0.02	28.20 ± 0.02	28.48 ± 0.02	36.49 ± 0.02	28.91 ± 0.02	32.53 ± 1.00	31.38	31.43	31.92	27.12 ± 0.04	31.32	32.00 ± 0.02	30.84	27.77 ± 0.02	24.00 ± 0.01	22.75 ± 0.01	26.42 ± 0.30
	Avg	28.69 ± 0.10	26.56 ± 0.08	28.95 ± 0.09	28.01 ± 0.05	10.64 ± 0.02	16.29 ± 0.02	29.62 ± 0.02	24.96 ± 0.02	27.63 ± 0.02	27.78 ± 0.02	27.58 ± 0.02	30.12 ± 0.02	26.12 ± 0.02	26.15 ± 0.02	26.44 ± 0.02	32.94 ± 0.02	26.79 ± 0.02	28.55 ± 0.05	25.88	25.83	26.71	23.79 ± 0.04	26.24	27.23 ± 0.03	26.15	24.18 ± 0.02	23.06 ± 0.01	21.79 ± 0.01	24.61 ± 0.28

PEMS04

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	21.15 ± 0.21	20.23 ± 0.20	22.27 ± 0.22	22.00 ± 0.22	15.31 ± 0.15	19.92 ± 0.19	15.81 ± 0.15	19.34 ± 0.19	19.74 ± 0.19	23.62 ± 0.23	21.20 ± 0.21	13.75 ± 0.13	20.31 ± 0.20	19.59 ± 0.19	21.19 ± 0.21	22.37 ± 0.22	10.59 ± 0.10	21.27 ± 0.21	17.83	17.60	19.32	17.78 ± 0.09	21.81	26.08 ± 0.30	20.63	17.86 ± 0.16	16.62 ± 0.16	16.17 ± 0.17	17.06 ± 0.05
	6	21.75 ± 0.22	22.06 ± 0.22	23.22 ± 0.23	22.97 ± 0.23	15.31 ± 0.15	21.35 ± 0.21	18.21 ± 0.20	20.78 ± 0.21	21.18 ± 0.21	24.90 ± 0.24	22.41 ± 0.22	16.61 ± 0.16	21.63 ± 0.21	21.04 ± 0.21	21.96 ± 0.22	23.63 ± 0.23	20.94 ± 0.20	23.12 ± 0.23	20.98	20.75	22.92	20.46 ± 0.09	26.14	30.01 ± 0.30	24.04	20.27 ± 0.19	19.33 ± 0.19	17.00 ± 0.17	18.64 ± 0.08
	12	24.38 ± 0.25	25.11 ± 0.26	26.97 ± 0.26	26.06 ± 0.25	15.31 ± 0.15	24.45 ± 0.24	21.12 ± 0.23	23.68 ± 0.24	24.16 ± 0.24	27.74 ± 0.27	25.55 ± 0.25	17.67 ± 0.17	24.64 ± 0.24	24.17 ± 0.24	24.86 ± 0.25	26.74 ± 0.26	26.91 ± 0.27	27.45 ± 0.27	28.71	28.27	30.08	25.86 ± 0.09	33.86	39.47 ± 0.09	30.91	25.23 ± 0.20	20.61 ± 0.20	18.42 ± 0.18	21.65 ± 0.15
	Avg	22.22 ± 0.22	22.45 ± 0.22	23.60 ± 0.23	23.41 ± 0.23	15.33 ± 0.15	21.64 ± 0.21	18.46 ± 0.21	21.01 ± 0.21	21.43 ± 0.21	25.16 ± 0.25	22.78 ± 0.22	14.02 ± 0.14	20.93 ± 0.20	21.33 ± 0.21	22.43 ± 0.22	23.99 ± 0.24	21.22 ± 0.21	23.58 ± 0.23	21.88	21.58	23.49	20.92 ± 0.09	26.72	31.16 ± 0.30	24.52	20.68 ± 0.19	19.38 ± 0.19	17.05 ± 0.17	18.84 ± 0.09
RMSE	3	32.49 ± 0.32	31.23 ± 0.31	35.57 ± 0.35	33.27 ± 0.33	15.75 ± 0.15	20.57 ± 0.20	14.99 ± 0.14	19.87 ± 0.19	20.41 ± 0.20	36.55 ± 0.36	34.87 ± 0.34	13.57 ± 0.13	20.86 ± 0.20	20.17 ± 0.20	23.49 ± 0.23	24.23 ± 0.24	10.24 ± 0.10	22.51 ± 0.22	29.20	28.79	30.99	29.58 ± 0.09	34.99	41.11 ± 0.50	32.99	29.40 ± 0.29	29.26 ± 0.29	26.28 ± 0.26	27.60 ± 0.07
	6	33.93 ± 0.34	34.42 ± 0.34	38.28 ± 0.38	35.06 ± 0.35	16.05 ± 0.16	23.18 ± 0.23	15.49 ± 0.15	22.43 ± 0.22	23.00 ± 0.23	38.80 ± 0.38	34.23 ± 0.34	14.58 ± 0.14	23.36 ± 0.23	22.80 ± 0.22	24.98 ± 0.25	25.16 ± 0.25	12.65 ± 0.12	25.90 ± 0.25	34.20	33.78	36.72	34.39 ± 0.09	41.80	47.11 ± 0.30	38.04	33.63 ± 0.30	30.61 ± 0.30	27.78 ± 0.27	30.25 ± 0.13
	12	38.46 ± 0.40	40.98 ± 0.42	42.75 ± 0.40	40.07 ± 0.38	17.60 ± 0.17	28.43 ± 0.28	15.38 ± 0.15	27.23 ± 0.27	27.99 ± 0.28	48.47 ± 0.48	39.70 ± 0.39	15.93 ± 0.15	38.50 ± 0.38	36.08 ± 0.36	39.67 ± 0.40	40.75 ± 0.40	12.57 ± 0.12	43.10 ± 0.43	47.13	46.16	48.32	43.96 ± 0.09	53.81	64.62 ± 2.50	48.84	42.50 ± 0.32	32.76 ± 0.32	30.12 ± 0.30	35.00 ± 0.25
	Avg	34.58 ± 0.34	34.96 ± 0.34	38.82 ± 0.38	35.69 ± 0.35	16.37 ± 0.16	23.58 ± 0.23	15.85 ± 0.15	22.73 ± 0.22	23.35 ± 0.23	39.17 ± 0.39	34.78 ± 0.34	14.80 ± 0.14	23.78 ± 0.23	23.21 ± 0.22	25.63 ± 0.25	25.65 ± 0.25	12.84 ± 0.12	26.54 ± 0.26	35.80	35.22	37.69	35.20 ± 0.09	42.74	49.27 ± 0.50	38.87	34.42 ± 0.30	30.64 ± 0.30	27.82 ± 0.27	30.50 ± 0.14
MAPE (%)	3	19.95 ± 0.17	17.57 ± 0.19	19.35 ± 0.18	18.96 ± 0.19	12.38 ± 0.12	17.87 ± 0.17	12.59 ± 0.12	15.67 ± 0.15	17.63 ± 0.17	21.16 ± 0.20	20.61 ± 0.20	11.87 ± 0.11	18.37 ± 0.18	17.74 ± 0.17	17.69 ± 0.17	15.18 ± 0.15	18.27 ± 0.18	17.68 ± 0.17	14.82	14.71	16.27	14.69 ± 0.09	16.45	17.91 ± 0.09	16.23	14.90 ± 0.16	16.44 ± 0.16	14.59 ± 0.14	15.51 ± 0.22
	6	20.17 ± 0.18	18.89 ± 0.19	19.76 ± 0.19	19.60 ± 0.20	12.26 ± 0.12	18.86 ± 0.18	12.67 ± 0.12	17.58 ± 0.17	18.53 ± 0.18	22.08 ± 0.22	21.21 ± 0.21	11.74 ± 0.11	19.14 ± 0.19	18.75 ± 0.18	18.18 ± 0.18	17.56 ± 0.17	19.11 ± 0.19	19.09 ± 0.19	17.01	16.93	18.84	16.33 ± 0.09	19.03	20.80 ± 0.09	18.75	16.48 ± 0.16	16.86 ± 0.16	15.09 ± 0.15	16.61 ± 0.28
	12	21.91 ± 0.22	22.29 ± 0.24	24.86 ± 0.21	21.79 ± 0.21	12.01 ± 0.12	20.34 ± 0.20	12.13 ± 0.12	19.79 ± 0.20	20.92 ± 0.24	24.23 ± 0.23	23.44 ± 0.22	12.12 ± 0.12	21.54 ± 0.21	21.24 ± 0.20	20.14 ± 0.21	21.05 ± 0.22	14.23 ± 0.14	22.99 ± 0.40	22.74	22.62	24.24	19.78 ± 0.09	23.86	26.17 ± 0.30	23.60	19.89 ± 0.17	17.90 ± 0.17	16.15 ± 0.16	18.89 ± 0.37
	Avg	20.53 ± 0.19	19.29 ± 0.20	20.15 ± 0.19	19.94 ± 0.20	12.25 ± 0.12	19.15 ± 0.19	12.93 ± 0.12	17.82 ± 0.17	18.83 ± 0.18	22.29 ± 0.22	21.54 ± 0.21	11.99 ± 0.11	19.49 ± 0.19	19.04 ± 0.18	18.51 ± 0.17	17.98 ± 0.17	19.36 ± 0.19	19.60 ± 0.19	17.72	17.62	19.32	16.65 ± 0.09	19.38	21.25 ± 0.09	19.11	16.78 ± 0.16	16.97 ± 0.16	15.18 ± 0.15	16.80 ± 0.29

PEMS05

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	11.22 ± 0.03	10.77 ± 0.03	10.31 ± 0.03	11.71 ± 0.03	12.31 ± 0.03	10.54 ± 0.03	14.56 ± 0.03	10.39 ± 0.03	10.54 ± 0.03	10.93 ± 0.03	10.73 ± 0.03	15.58 ± 0.03	10.58 ± 0.03	10.45 ± 0.03	10.03 ± 0.03	10.98 ± 0.03	10.69 ± 0.03	10.00 ± 0.03	10.86	10.72	11.20	10.44 ± 0.03	12.13	13.76 ± 0.03	11.56	10.39 ± 0.03	10.88 ± 0.03	9.31 ± 0.03	10.80 ± 0.03
	6	11.49 ± 0.03	10.55 ± 0.03	10.79 ± 0.03	12.34 ± 0.03	12.77 ± 0.03	11.20 ± 0.03	16.76 ± 0.03	11.00 ± 0.03	10.21 ± 0.03	11.55 ± 0.03	11.36 ± 0.03	16.21 ± 0.03	11.21 ± 0.03	10.11 ± 0.03	10.48 ± 0.03	12.44 ± 0.03	11.30 ± 0.03	10.96 ± 0.03	12.33	12.23	12.84	11.62 ± 0.03	14.04	15.48 ± 0.03	12.97	11.46 ± 0.03	12.12 ± 0.03	9.61 ± 0.03	10.40 ± 0.03
	12	12.63 ± 0.03	10.21 ± 0.03	10.04 ± 0.03	14.10 ± 0.03	16.01 ± 0.03	12.57 ± 0.03	18.26 ± 0.03	12.24 ± 0.03	12.57 ± 0.03	12.87 ± 0.03	12.72 ± 0.03	17.64 ± 0.03	12.55 ± 0.03	10.48 ± 0.03	10.72 ± 0.03	10.56 ± 0.03	12.62 ± 0.03	14.06 ± 0.03	16.08	15.95	16.14	14.09 ± 0.03	17.55	18.96 ± 0.03	16.18	13.73 ± 0.03	12.60 ± 0.03	10.12 ± 0.03	11.57 ± 0.03
	Avg	11.69 ± 0.03	10.69 ± 0.03	10.94 ± 0.03	12.57 ± 0.03	12.94 ± 0.03	11.31 ± 0.03	16.98 ± 0.03	11.10 ± 0.03	10.32 ± 0.03	11.67 ± 0.03	11.48 ± 0.03	16.35 ± 0.03	11.33 ± 0.03	10.22 ± 0.03	10.64 ± 0.03	12.57 ± 0.03	11.42 ± 0.03	12.16 ± 0.03	12.79	12.67	13.11	11.85 ± 0.03	14.34	15.75 ± 0.03	13.26	11.66 ± 0.03	12.16 ± 0.03	9.63 ± 0.03	10.48 ± 0.03
RMSE	3	17.36 ± 0.06	15.92 ± 0.06	16.29 ± 0.06	17.96 ± 0.06	18.60 ± 0.06	15.59 ± 0.06	28.55 ± 0.06	15.33 ± 0.06	16.61 ± 0.06	17.14 ± 0.06	16.84 ± 0.06	22.45 ± 0.06	15.58 ± 0.06	16.46 ± 0.06	17.64 ± 0.06	18.03 ± 0.06	15.69 ± 0.06	17.33 ± 0.06	17.39	17.14	17.78	16.98 ± 0.06	19.54	21.76 ± 0.06	18.39	16.77 ± 0.06	16.54 ± 0.06	14.82 ± 0.06	15.64 ± 0.06
	6	18.03 ± 0.06	16.31 ± 0.06	19.20 ± 0.06	19.12 ± 0.06	14.52 ± 0.06	17.84 ± 0.06	26.94 ± 0.06	17.49 ± 0.06	17.87 ± 0.06	18.33 ± 0.06	18.07 ± 0.06	23.58 ± 0.06	17.79 ± 0.06	17.71 ± 0.06	18.43 ± 0.06	18.81 ± 0.06	17.86 ± 0.06	19.09 ± 0.06	19.96	19.78	20.61	19.44 ± 0.06	22.68	24.70 ± 0.06	20.68	18.85 ± 0.06	16.02 ± 0.06	15.38 ± 0.06	16.75 ± 0.06
	12	20.16 ± 0.06	20.07 ± 0.06	21.40 ± 0.06	22.14 ± 0.06	16.72 ± 0.06	20.30 ± 0.06	30.76 ± 0.06	19.68 ± 0.06	20.32 ± 0.06	20.68 ± 0.06	20.57 ± 0.06	25.93 ± 0.06	20.19 ± 0.06	20.17 ± 0.06	20.58 ± 0.06	20.61 ± 0.06	20.20 ± 0.06	22.73 ± 0.06	26.60	26.27	26.26	24.56 ± 0.06	28.35	30.31 ± 0.06	25.89	23.50 ± 0.06	16.89 ± 0.06	16.29 ± 0.06	18.77 ± 0.13
	Avg	18.34 ± 0.06	16.52 ± 0.06	19.43 ± 0.06	19.49 ± 0.06	14.77 ± 0.06	18.03 ± 0.06	26.35 ± 0.06	17.63 ± 0.06	18.04 ± 0.06	18.50 ± 0.06	18.26 ± 0.06	23.77 ± 0.06	17.97 ± 0.06	17.89 ± 0.06	18.69 ± 0.06	19.00 ± 0.06	18.04 ± 0.06	19.40 ± 0.06	20.79	20.54	21.07	19.92 ± 0.06	23.12	25.07 ± 0.06	21.14	19.31 ± 0.06	16.07 ± 0.06	15.40 ± 0.06	16.85 ± 0.08
MAPE (%)	3	26.28 ± 0.08	28.31 ± 0.08	26.81 ± 0.08	26.60 ± 0.08	14.78 ± 0.08	23.06 ± 0.08	27.15 ± 0.08	22.28 ± 0.08	23.18 ± 0.08	23.88 ± 0.08	23.69 ± 0.08	33.98 ± 0.08	23.90 ± 0.08	22.97 ± 0.08	23.40 ± 0.08	33.91 ± 0.08	23.53 ± 0.08	23.50 ± 0.08	22.07	21.96	23.23	20.62 ± 0.08	22.62	23.96 ± 0.08	22.69	21.26 ± 0.08	22.26 ± 0.08	20.24 ± 0.08	20.44 ± 0.08
	6	26.59 ± 0.08	24.92 ± 0.08	26.46 ± 0.08	26.73 ± 0.08	15.48 ± 0.08	24.15 ± 0.08	28.67 ± 0.08	23.24 ± 0.08	24.29 ± 0.08	24.84 ± 0.08	24.82 ± 0.08	60.44 ± 0.08	24.93 ± 0.08	24.05 ± 0.08	24.20 ± 0.08	35.01 ± 0.08	24.65 ± 0.08	25.54 ± 0.08	24.65	24.72	26.15	22.15 ± 0.08	25.53	27.63 ± 0.08	25.33	22.97 ± 0.08	22.67 ± 0.08	20.74 ± 0.08	21.32 ± 0.03
	12	28.62 ± 0.08	28.72 ± 0.08	28.84 ± 0.08	30.08 ± 0.08	18.51 ± 0.08	26.77 ± 0.08	32.35 ± 0.08	25.61 ± 0.08	26.89 ± 0.08	27.37 ± 0.08	27.67 ± 0.08	63.79 ± 0.08	27.61 ± 0.08	26.67 ± 0.08	26.50 ± 0.08	38.11 ± 0.08	27.28 ± 0.08	30.73 ± 0.08	31.79	32.01	32.54	25.58 ± 0.08	31.30	33.24 ± 0.08	31.29	26.70 ± 0.08	26.86 ± 0.08	21.83 ± 0.08	23.25 ± 0.04
	Avg	26.98 ± 0.08	25.31 ± 0.08	26.83 ± 0.08	27.19 ± 0.08	16.10 ± 0.08	24.42 ± 0.08	29.06 ± 0.08	23.49 ± 0.08	24.56 ± 0.08	25.13 ± 0.08	25.11 ± 0.08	61.09 ± 0.08	25.25 ± 0.08	24.33 ± 0.08	24.51 ± 0.08	35.42 ± 0.08	24.92 ± 0.08	25.15 ± 0.08	25.60	25.64	26.75	22.50 ± 0.08	25.96	27.92 ± 0.08	25.93	23.30 ± 0.08	22.83 ± 0.08	20.84 ± 0.08	21.49 ± 0.03

PEMS06

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAE-Forn	STAE-Our	A2TTA
MAE	3	14.29 ± 0.1	14.25 ± 0.1	14.75 ± 0.1	14.91 ± 0.1	13.18 ± 0.1	13.39 ± 0.1	15.65 ± 0.1	13.10 ± 0.1	13.47 ± 0.1	17.23 ± 0.1	13.86 ± 0.1	19.38 ± 0.1	13.53 ± 0.1	13.28 ± 0.1	14.65 ± 0.1	15.70 ± 0.1	14.19 ± 0.1	14.86 ± 0.1	13.18	13.03	13.60	12.46 ± 0.0	14.70	17.57 ± 0.4	14.03	12.68 ± 0.1	12.73 ± 0.1	11.82 ± 0.1	12.05 ± 0.10
	6	14.60 ± 0.1	15.36 ± 0.1	15.33 ± 0.1	15.92 ± 0.1	13.77 ± 0.1	14.24 ± 0.1	17.53 ± 0.1	13.91 ± 0.1	14.33 ± 0.1	17.90 ± 0.1	14.75 ± 0.1	20.83 ± 0.1	14.30 ± 0.1	14.13 ± 0.1	15.20 ± 0.1	15.52 ± 0.1	15.02 ± 0.1	16.16 ± 0.1	15.19	15.11	15.79	14.02 ± 0.0	17.33	19.73 ± 0.4	16.01	14.08 ± 0.1	13.09 ± 0.1	12.24 ± 0.1	12.89 ± 0.14
	12	16.11 ± 0.1	17.62 ± 0.1	16.90 ± 0.1	18.24 ± 0.1	19.36 ± 0.1	15.99 ± 0.1	19.23 ± 0.1	16.54 ± 0.1	16.09 ± 0.1	19.37 ± 0.1	16.67 ± 0.1	23.88 ± 0.1	15.99 ± 0.1	15.88 ± 0.1	16.77 ± 0.1	18.34 ± 0.1	16.83 ± 0.1	18.94 ± 0.1	20.01	19.88	20.01	17.21 ± 0.0	22.01	24.25 ± 0.3	20.28	16.91 ± 0.1	13.78 ± 0.1	12.97 ± 0.1	14.46 ± 0.20
	Avg	14.87 ± 0.1	15.52 ± 0.1	15.52 ± 0.1	16.16 ± 0.1	13.66 ± 0.1	14.37 ± 0.1	17.64 ± 0.1	14.02 ± 0.1	14.46 ± 0.1	18.02 ± 0.1	14.90 ± 0.1	21.10 ± 0.1	14.44 ± 0.1	14.25 ± 0.1	15.39 ± 0.1	15.70 ± 0.1	15.18 ± 0.1	16.40 ± 0.1	15.71	15.59	16.08	14.28 ± 0.0	17.67	20.04 ± 0.4	16.33	14.28 ± 0.1	13.13 ± 0.1	12.27 ± 0.1	12.97 ± 0.14
RMSE	3	22.13 ± 0.1	24.74 ± 0.1	23.99 ± 0.1	22.66 ± 0.1	19.67 ± 0.1	20.57 ± 0.1	36.24 ± 0.1	20.00 ± 0.1	20.57 ± 0.1	27.60 ± 0.1	21.19 ± 0.1	28.19 ± 0.1	20.82 ± 0.1	20.39 ± 0.1	23.12 ± 0.1	23.42 ± 0.1	24.56 ± 0.1	22.68 ± 0.1	20.98	20.74	21.65	20.03 ± 0.0	23.69	27.96 ± 0.6	22.24	20.34 ± 0.1	19.15 ± 0.1	17.87 ± 0.1	18.65 ± 0.19
	6	22.84 ± 0.1	23.84 ± 0.1	25.08 ± 0.1	24.49 ± 0.1	19.00 ± 0.1	22.23 ± 0.1	38.11 ± 0.1	21.54 ± 0.1	22.27 ± 0.1	28.90 ± 0.1	22.98 ± 0.1	30.83 ± 0.1	22.35 ± 0.1	22.06 ± 0.1	24.03 ± 0.1	24.72 ± 0.1	23.16 ± 0.1	25.24 ± 0.1	24.99	24.87	25.77	23.64 ± 0.0	28.35	32.03 ± 0.8	25.81	23.30 ± 0.1	19.84 ± 0.1	18.65 ± 0.1	20.22 ± 0.28
	12	25.62 ± 0.1	27.71 ± 0.1	27.79 ± 0.1	28.42 ± 0.1	20.02 ± 0.1	25.34 ± 0.1	41.47 ± 0.1	24.41 ± 0.1	25.39 ± 0.1	31.36 ± 0.1	26.49 ± 0.1	35.07 ± 0.1	26.35 ± 0.1	26.18 ± 0.1	25.62 ± 0.1	27.54 ± 0.1	26.36 ± 0.1	30.13 ± 0.1	33.98	33.64	33.36	30.70 ± 0.0	36.75	44.91 ± 8.7	33.00	29.32 ± 0.1	20.03 ± 0.1	19.94 ± 0.1	23.06 ± 0.38
	Avg	23.29 ± 0.1	24.04 ± 0.1	25.38 ± 0.1	24.84 ± 0.1	19.96 ± 0.1	22.39 ± 0.1	38.29 ± 0.1	21.68 ± 0.1	22.41 ± 0.1	29.03 ± 0.1	23.19 ± 0.1	31.23 ± 0.1	22.53 ± 0.1	22.21 ± 0.1	24.34 ± 0.1	24.97 ± 0.1	23.39 ± 0.1	25.55 ± 0.1	25.88	25.65	26.22	24.21 ± 0.0	28.85	32.90 ± 1.1	26.23	23.75 ± 0.1	19.88 ± 0.1	18.68 ± 0.1	20.33 ± 0.27
MAPE (%)	3	24.10 ± 0.1	21.76 ± 0.1	23.21 ± 0.1	23.31 ± 0.1	19.39 ± 0.1	21.38 ± 0.1	31.34 ± 0.1	20.89 ± 0.1	21.45 ± 0.1	25.89 ± 0.1	23.18 ± 0.1	34.95 ± 0.1	21.52 ± 0.1	21.25 ± 0.1	21.75 ± 0.1	20.95 ± 0.1	22.76 ± 0.1	22.34 ± 0.1	19.47	19.36	20.16	18.36 ± 0.0	19.80	21.95 ± 0.3	19.90	18.89 ± 0.1	19.81 ± 0.1	18.86 ± 0.1	19.59 ± 0.31
	6	24.33 ± 0.1	23.07 ± 0.1	23.76 ± 0.1	24.42 ± 0.1	19.52 ± 0.1	22.31 ± 0.1	32.02 ± 0.1	21.73 ± 0.1	22.24 ± 0.1	26.69 ± 0.1	23.91 ± 0.1	36.41 ± 0.1	22.36 ± 0.1	22.19 ± 0.1	22.48 ± 0.1	22.23 ± 0.1	23.77 ± 0.1	24.02 ± 0.1	21.56	21.61	22.53	19.79 ± 0.0	22.19	24.49 ± 0.3	22.05	20.42 ± 0.1	20.23 ± 0.1	19.28 ± 0.1	20.45 ± 0.37
	12	25.99 ± 0.1	26.28 ± 0.1	26.75 ± 0.1	27.32 ± 0.1	19.07 ± 0.1	24.56 ± 0.1	33.58 ± 0.1	23.69 ± 0.1	24.14 ± 0.1	28.62 ± 0.1	25.98 ± 0.1	39.24 ± 0.1	24.53 ± 0.1	24.43 ± 0.1	24.45 ± 0.1	23.38 ± 0.1	26.08 ± 0.1	23.21 ± 0.1	27.09	27.27	27.51	22.86 ± 0.0	26.87	28.81 ± 0.2	26.63	23.73 ± 0.1	20.03 ± 0.1	20.08 ± 0.1	22.18 ± 0.58
	Avg	24.66 ± 0.1	23.41 ± 0.1	24.06 ± 0.1	24.76 ± 0.1	19.40 ± 0.1	22.54 ± 0.1	32.16 ± 0.1	21.93 ± 0.1	22.43 ± 0.1	26.88 ± 0.1	23.14 ± 0.1	36.63 ± 0.1	22.60 ± 0.1	22.42 ± 0.1	22.72 ± 0.1	22.58 ± 0.1	24.00 ± 0.1	24.49 ± 0.1	22.24	22.27	22.96	20.06 ± 0.0	22.54	24.70 ± 0.2	22.46	20.71 ± 0.1	20.28 ± 0.1	19.33 ± 0.1	20.58 ± 0.40

PEMS07

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	18.59 ± 0.07	17.59 ± 0.06	19.66 ± 0.08	18.38 ± 0.07	140.22 ± 216.53	± 0.21	15.47 ± 0.15	15.22 ± 0.15	15.12 ± 0.17	17.90 ± 15.62	± 0.17	17.21 ± 0.15	16.37 ± 0.16	16.24 ± 0.18	18.02 ± 0.23	18.09 ± 0.19	17.17 ± 0.17	17.63 ± 0.00	15.37	15.19	15.97	14.96 ± 0.00	18.20	20.28 ± 0.02	17.22	14.98 ± 0.01	14.67 ± 0.13	13.23 ± 0.04	14.13 ± 0.07
	6	18.93 ± 0.09	19.22 ± 0.20	20.25 ± 0.19	19.21 ± 0.40	10.32 ± 117.83	± 0.22	17.53 ± 0.17	17.40 ± 0.19	19.09 ± 116.93	± 0.18	18.14 ± 0.17	17.64 ± 0.17	17.55 ± 0.18	18.94 ± 0.24	17.77 ± 118.44	± 0.19	19.49 ± 0.00	18.26	18.03	19.10	17.51 ± 0.00	21.91	23.40 ± 0.01	20.19	17.24 ± 0.06	16.29 ± 0.13	13.98 ± 0.05	15.52 ± 0.08	
	12	21.19 ± 0.22	22.81 ± 0.22	22.49 ± 0.21	21.54 ± 0.40	10.46 ± 220.40	± 0.24	19.94 ± 0.19	19.79 ± 0.21	21.32 ± 119.34	± 0.19	19.99 ± 0.20	20.09 ± 0.20	20.14 ± 0.21	21.55 ± 0.28	21.54 ± 121.02	± 0.23	23.49 ± 0.00	25.04	24.47	25.20	22.33 ± 0.00	28.49	29.63 ± 0.01	26.07	21.56 ± 0.06	16.32 ± 0.15	15.14 ± 0.08	18.03 ± 0.10	
	Avg	19.39 ± 0.19	19.56 ± 0.20	20.62 ± 0.19	19.51 ± 0.40	10.30 ± 218.02	± 0.22	17.67 ± 0.17	17.54 ± 0.19	19.23 ± 117.07	± 0.18	18.26 ± 0.17	17.81 ± 0.17	17.74 ± 0.19	19.27 ± 0.25	18.17 ± 118.64	± 0.19	19.85 ± 0.00	18.98	18.67	19.55	17.86 ± 0.00	22.38	23.89 ± 0.01	20.58	17.53 ± 0.06	16.32 ± 0.13	13.99 ± 0.05	15.65 ± 0.08	
RMSE	3	30.83 ± 0.29	29.38 ± 0.35	35.00 ± 0.30	30.15 ± 152.97	± 227.82	± 0.33	28.25 ± 0.27	27.41 ± 0.27	27.28 ± 0.29	29.96 ± 226.60	± 128.38	± 0.27	27.62 ± 0.27	27.46 ± 0.29	29.94 ± 0.35	25.09 ± 128.42	± 0.29	29.63 ± 0.00	27.86	27.55	28.60	27.46 ± 0.00	32.69	36.91 ± 0.01	30.43	27.18 ± 0.06	26.56 ± 0.23	23.80 ± 0.05	25.08 ± 0.06
	6	31.89 ± 0.32	32.54 ± 0.37	37.21 ± 0.31	31.78 ± 154.08	± 130.46	± 0.35	29.59 ± 0.29	29.94 ± 0.29	29.84 ± 0.32	30.30 ± 229.14	± 130.27	± 0.30	30.13 ± 0.30	30.10 ± 0.31	31.73 ± 0.38	38.85 ± 130.90	± 0.33	33.33 ± 0.00	33.35	32.91	34.44	32.80 ± 0.00	39.61	42.56 ± 0.01	35.69	31.79 ± 0.06	26.92 ± 0.25	25.37 ± 0.07	27.72 ± 0.09
	12	36.27 ± 0.39	39.02 ± 0.41	41.01 ± 0.36	36.84 ± 155.87	± 135.29	± 0.39	38.86 ± 0.34	34.34 ± 0.34	29.86 ± 0.35	33.55 ± 138.61	± 0.34	34.67 ± 0.34	34.96 ± 0.35	35.28 ± 0.45	35.39 ± 235.56	± 0.40	40.75 ± 0.01	46.69	45.31	46.03	42.92 ± 0.00	52.40	59.72 ± 6.8	46.50	40.60 ± 0.06	28.97 ± 0.27	27.61 ± 0.02	32.13 ± 0.14	
	Avg	32.62 ± 0.33	33.05 ± 0.37	37.75 ± 0.32	32.22 ± 154.10	± 130.73	± 0.35	30.82 ± 0.30	30.13 ± 0.30	30.04 ± 0.32	32.52 ± 229.33	± 130.41	± 0.30	30.37 ± 0.30	30.37 ± 0.32	32.22 ± 0.39	35.57 ± 131.19	± 0.33	33.90 ± 0.00	34.86	34.18	35.34	33.57 ± 0.00	40.57	44.62 ± 1.2	36.42	32.42 ± 0.06	26.93 ± 0.25	25.34 ± 0.07	27.87 ± 0.09
MAPE (%)	3	30.48 ± 0.27	27.98 ± 0.31	31.59 ± 0.30	30.73 ± 250.26	± 226.51	± 0.34	24.13 ± 0.25	25.57 ± 0.25	25.51 ± 0.26	26.75 ± 124.19	± 327.34	± 0.25	25.71 ± 0.26	26.27 ± 0.32	24.44 ± 0.55	25.33 ± 0.22	29.41 ± 0.27	27.44 ± 0.01	19.45	19.33	20.30	18.88 ± 0.00	20.43	21.13 ± 0.00	20.34	18.90 ± 0.02	20.97 ± 0.19	19.20 ± 0.02	20.61 ± 0.72
	6	30.62 ± 0.29	29.47 ± 0.31	31.86 ± 0.30	30.46 ± 249.24	± 227.39	± 0.34	24.81 ± 0.26	26.44 ± 0.26	26.43 ± 0.27	27.64 ± 125.38	± 328.43	± 0.25	26.65 ± 0.27	27.16 ± 0.33	23.59 ± 0.58	28.00 ± 0.23	30.46 ± 0.29	29.34 ± 0.01	22.00	21.88	23.14	20.91 ± 0.00	23.33	24.52 ± 0.01	23.12	20.85 ± 0.02	20.31 ± 0.19	19.64 ± 0.02	21.73 ± 0.67
	12	32.67 ± 0.33	34.46 ± 0.34	34.54 ± 0.36	36.85 ± 247.91	± 329.62	± 0.36	28.82 ± 0.28	28.63 ± 0.28	28.65 ± 0.29	29.78 ± 128.11	± 331.10	± 0.28	28.92 ± 0.29	29.40 ± 0.37	27.31 ± 0.54	24.27 ± 0.43	23.08 ± 0.34	24.30 ± 0.01	28.62	28.40	29.16	24.96 ± 0.00	28.97	29.96 ± 0.00	28.69	24.87 ± 0.02	22.17 ± 0.20	20.60 ± 0.02	24.19 ± 0.59
	Avg	31.09 ± 0.29	29.95 ± 0.32	32.43 ± 0.30	30.81 ± 249.24	± 227.64	± 0.35	25.08 ± 0.26	26.68 ± 0.26	26.67 ± 0.27	27.85 ± 125.66	± 328.73	± 0.25	26.88 ± 0.27	27.40 ± 0.34	23.13 ± 0.58	28.70 ± 0.23	30.76 ± 0.29	29.94 ± 0.01	22.81	22.66	23.68	21.24 ± 0.00	23.73	24.80 ± 0.00	23.57	21.17 ± 0.02	20.39 ± 0.19	19.72 ± 0.02	21.98 ± 0.67

PEMS08

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	14.86 ± 0.03	13.63 ± 0.01	16.01 ± 0.04	14.62 ± 0.05	2.44 ± 0.21	3.13 ± 0.13	16.14 ± 0.02	12.76 ± 0.02	12.85 ± 0.01	14.58 ± 0.01	18.32 ± 0.01	18.86 ± 0.01	12.98 ± 0.01	12.95 ± 0.01	14.23 ± 0.01	17.19 ± 0.01	13.42 ± 0.01	16.79 ± 0.01	13.40	13.26	13.94	12.77 ± 0.01	15.65	17.49 ± 0.01	14.90	13.02 ± 0.01	17.99 ± 0.01	11.28 ± 0.02	12.04 ± 0.03
	6	15.17 ± 0.04	14.97 ± 0.01	16.60 ± 0.05	15.39 ± 0.05	2.24 ± 0.21	4.18 ± 0.17	17.16 ± 0.01	16.80 ± 0.01	13.91 ± 0.01	15.47 ± 0.01	14.31 ± 0.01	14.66 ± 0.01	14.02 ± 0.01	14.00 ± 0.01	14.83 ± 0.01	18.07 ± 0.01	14.44 ± 0.01	16.28 ± 0.01	15.75	15.62	16.49	14.62 ± 0.01	18.77	19.98 ± 0.01	17.30	14.75 ± 0.01	17.47 ± 0.01	11.83 ± 0.03	13.11 ± 0.04
	12	16.88 ± 0.07	17.84 ± 0.01	17.43 ± 0.07	17.61 ± 0.05	2.07 ± 0.31	6.27 ± 0.19	19.13 ± 0.01	15.75 ± 0.01	15.90 ± 0.01	17.19 ± 0.01	16.32 ± 0.01	16.25 ± 0.01	16.00 ± 0.01	16.10 ± 0.01	16.83 ± 0.01	20.27 ± 0.01	16.48 ± 0.01	18.53 ± 0.01	21.43	21.09	21.51	18.18 ± 0.01	24.43	25.47 ± 0.01	22.15	18.10 ± 0.01	18.32 ± 0.01	12.73 ± 0.05	15.04 ± 0.05
	Avg	15.50 ± 0.01	15.22 ± 0.01	16.86 ± 0.01	15.69 ± 0.02	2.25 ± 0.21	4.34 ± 0.17	17.30 ± 0.01	16.92 ± 0.01	14.04 ± 0.01	15.58 ± 0.01	14.47 ± 0.01	14.76 ± 0.01	14.15 ± 0.01	14.16 ± 0.01	16.12 ± 0.01	18.33 ± 0.01	14.60 ± 0.01	16.58 ± 0.01	16.39	16.19	16.87	14.90 ± 0.01	19.21	20.52 ± 0.01	17.64	14.99 ± 0.01	17.52 ± 0.01	11.85 ± 0.03	13.21 ± 0.04
RMSE	3	23.73 ± 0.02	22.36 ± 0.02	26.25 ± 0.03	23.28 ± 0.05	1.63 ± 0.22	1.49 ± 0.24	24.62 ± 0.02	20.01 ± 0.02	21.09 ± 0.02	23.27 ± 0.02	22.64 ± 0.02	21.99 ± 0.02	21.29 ± 0.02	21.26 ± 0.02	24.03 ± 0.02	26.14 ± 0.02	24.79 ± 0.02	22.74 ± 0.02	22.55	22.30	23.20	21.87 ± 0.02	26.33	29.66 ± 0.05	24.72	22.10 ± 0.01	19.83 ± 0.01	18.90 ± 0.02	20.16 ± 0.05
	6	24.70 ± 0.02	24.88 ± 0.02	27.48 ± 0.02	24.83 ± 0.05	2.11 ± 0.32	6.63 ± 0.26	26.65 ± 0.02	23.05 ± 0.02	23.19 ± 0.02	25.08 ± 0.02	26.66 ± 0.02	28.65 ± 0.02	23.35 ± 0.02	26.41 ± 0.02	26.24 ± 0.02	27.58 ± 0.02	23.77 ± 0.02	25.65 ± 0.02	26.85	26.57	27.72	25.66 ± 0.02	31.97	33.94 ± 0.03	28.76	25.56 ± 0.02	20.82 ± 0.02	20.01 ± 0.02	22.22 ± 0.07
	12	28.02 ± 0.03	29.80 ± 0.03	30.67 ± 0.03	28.68 ± 0.03	3.24 ± 0.27	7.52 ± 0.30	30.31 ± 0.02	26.53 ± 0.02	26.81 ± 0.02	28.33 ± 0.02	27.41 ± 0.02	26.49 ± 0.02	27.00 ± 0.02	27.31 ± 0.02	28.72 ± 0.02	31.18 ± 0.02	27.42 ± 0.02	31.37 ± 0.02	37.17	36.33	36.53	32.74 ± 0.02	41.51	47.49 ± 0.13	37.04	31.97 ± 0.02	22.44 ± 0.02	21.69 ± 0.05	25.64 ± 0.10
	Avg	25.20 ± 0.02	25.23 ± 0.02	27.85 ± 0.02	25.27 ± 0.02	2.21 ± 0.32	3.85 ± 0.26	26.86 ± 0.02	23.20 ± 0.02	23.35 ± 0.02	25.25 ± 0.02	26.89 ± 0.02	29.76 ± 0.02	23.54 ± 0.02	26.63 ± 0.02	26.69 ± 0.02	28.00 ± 0.02	23.99 ± 0.02	26.07 ± 0.02	28.01	27.58	28.38	26.18 ± 0.02	32.55	35.46 ± 0.03	29.31	25.97 ± 0.02	20.88 ± 0.02	20.02 ± 0.02	22.34 ± 0.07
MAPE (%)	3	22.82 ± 0.01	19.07 ± 0.01	24.25 ± 0.02	20.76 ± 0.05	7.52 ± 0.11	9.25 ± 0.19	22.79 ± 0.01	18.05 ± 0.01	19.62 ± 0.01	24.16 ± 0.02	20.30 ± 0.02	26.19 ± 0.01	18.63 ± 0.01	19.13 ± 0.01	19.38 ± 0.01	41.36 ± 0.02	20.82 ± 0.01	18.96 ± 0.04	17.10	17.01	17.93	16.36 ± 0.01	18.02	18.90 ± 0.01	17.86	16.69 ± 0.01	17.16 ± 0.01	16.15 ± 0.06	16.65 ± 0.12
	6	22.94 ± 0.01	20.47 ± 0.01	21.71 ± 0.01	21.51 ± 0.05	6.94 ± 0.20	10.33 ± 0.23	23.83 ± 0.01	19.02 ± 0.02	20.70 ± 0.02	22.07 ± 0.02	24.30 ± 0.02	26.35 ± 0.01	19.69 ± 0.02	20.20 ± 0.01	19.98 ± 0.02	42.87 ± 0.02	21.81 ± 0.02	20.66 ± 0.04	19.36	19.33	20.46	17.97 ± 0.01	20.63	21.71 ± 0.01	20.31	18.33 ± 0.01	17.53 ± 0.01	16.55 ± 0.07	17.59 ± 0.10
	12	24.58 ± 0.02	28.92 ± 0.02	24.00 ± 0.02	24.07 ± 0.05	6.54 ± 0.22	7.74 ± 0.26	26.26 ± 0.02	24.37 ± 0.02	23.18 ± 0.02	24.14 ± 0.02	23.62 ± 0.02	27.80 ± 0.02	22.19 ± 0.02	22.61 ± 0.02	22.15 ± 0.04	46.66 ± 0.02	24.16 ± 0.02	24.95 ± 0.04	25.40	25.32	25.89	21.27 ± 0.01	25.69	26.68 ± 0.01	25.19	21.82 ± 0.01	18.49 ± 0.01	17.47 ± 0.09	19.71 ± 0.09
	Avg	23.31 ± 0.01	20.85 ± 0.01	22.13 ± 0.01	21.90 ± 0.05	7.09 ± 0.20	10.58 ± 0.24	24.09 ± 0.01	19.28 ± 0.02	20.96 ± 0.02	22.26 ± 0.02	24.55 ± 0.02	26.61 ± 0.01	19.96 ± 0.02	20.45 ± 0.02	20.33 ± 0.04	43.33 ± 0.02	22.07 ± 0.02	21.16 ± 0.04	20.13	20.06	20.96	18.26 ± 0.01	21.00	22.04 ± 0.01	20.70	18.63 ± 0.01	17.66 ± 0.01	16.65 ± 0.07	17.81 ± 0.10

PEMS10

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	11.81 ± 0.03	11.46 ± 0.03	10.82 ± 0.03	10.87 ± 0.03	11.21 ± 0.03	11.09 ± 0.03	10.49 ± 0.03	10.86 ± 0.03	10.14 ± 0.03	10.90 ± 0.03	11.15 ± 0.03	10.84 ± 0.03	11.05 ± 0.03	10.01 ± 0.03	10.85 ± 0.03	10.13 ± 0.03	11.13 ± 0.03	10.73 ± 0.03	11.64	11.46	11.82	11.04 ± 0.03	13.03	14.44 ± 0.03	12.56	11.24 ± 0.03	10.09 ± 0.03	10.65 ± 0.03	10.36 ± 0.03
	6	12.06 ± 0.03	12.32 ± 0.03	10.29 ± 0.03	10.38 ± 0.03	11.43 ± 0.03	11.74 ± 0.03	10.23 ± 0.03	11.50 ± 0.03	11.82 ± 0.03	10.51 ± 0.03	11.81 ± 0.03	10.48 ± 0.03	11.68 ± 0.03	10.65 ± 0.03	10.36 ± 0.03	10.62 ± 0.03	11.74 ± 0.03	10.69 ± 0.03	13.06	12.93	13.38	12.10 ± 0.03	15.01	16.15 ± 0.03	13.93	12.30 ± 0.03	10.32 ± 0.03	10.91 ± 0.03	10.97 ± 0.03
	12	13.19 ± 0.03	14.05 ± 0.03	10.48 ± 0.03	10.70 ± 0.03	11.53 ± 0.03	12.06 ± 0.03	10.70 ± 0.03	12.76 ± 0.03	13.18 ± 0.03	10.72 ± 0.03	10.14 ± 0.03	10.87 ± 0.03	12.97 ± 0.03	10.97 ± 0.03	10.73 ± 0.03	10.48 ± 0.03	12.95 ± 0.03	10.67 ± 0.03	16.56	16.37	16.49	14.17 ± 0.03	18.64	19.70 ± 0.03	16.97	14.30 ± 0.03	10.73 ± 0.03	10.34 ± 0.03	10.16 ± 0.03
	Avg	12.26 ± 0.03	12.45 ± 0.03	10.43 ± 0.03	10.54 ± 0.03	11.39 ± 0.03	11.85 ± 0.03	10.35 ± 0.03	11.60 ± 0.03	10.93 ± 0.03	10.61 ± 0.03	11.92 ± 0.03	10.60 ± 0.03	11.79 ± 0.03	10.76 ± 0.03	10.53 ± 0.03	10.75 ± 0.03	11.83 ± 0.03	10.86 ± 0.03	13.47	13.30	13.63	12.27 ± 0.03	15.31	16.45 ± 0.03	14.18	12.43 ± 0.03	10.34 ± 0.03	10.92 ± 0.03	10.06 ± 0.03
RMSE	3	19.17 ± 0.06	19.03 ± 0.06	20.39 ± 0.06	19.21 ± 0.06	23.86 ± 0.06	23.21 ± 0.06	18.96 ± 0.06	17.84 ± 0.06	18.31 ± 0.06	20.00 ± 0.06	18.35 ± 0.06	19.71 ± 0.06	18.17 ± 0.06	18.08 ± 0.06	19.33 ± 0.06	20.78 ± 0.06	18.33 ± 0.06	19.35 ± 0.06	19.62	19.29	19.75	18.67 ± 0.06	21.83	23.87 ± 0.06	20.99	19.00 ± 0.06	16.69 ± 0.06	16.15 ± 0.06	17.27 ± 0.06
	6	19.79 ± 0.09	20.59 ± 0.09	20.24 ± 0.09	20.20 ± 0.09	24.50 ± 0.09	19.49 ± 0.09	20.39 ± 0.09	19.06 ± 0.09	19.58 ± 0.09	21.13 ± 0.09	19.63 ± 0.09	20.94 ± 0.09	19.40 ± 0.09	19.35 ± 0.09	20.30 ± 0.09	20.60 ± 0.09	19.49 ± 0.09	21.11 ± 0.09	22.12	21.87	22.48	20.80 ± 0.09	25.46	28.17 ± 0.09	23.25	21.08 ± 0.09	17.17 ± 0.09	16.66 ± 0.09	18.40 ± 0.09
	12	21.96 ± 0.12	23.58 ± 0.12	20.35 ± 0.12	22.52 ± 0.12	25.41 ± 0.12	24.94 ± 0.12	20.10 ± 0.12	21.39 ± 0.12	22.03 ± 0.12	23.31 ± 0.12	22.08 ± 0.12	23.29 ± 0.12	21.80 ± 0.12	21.80 ± 0.12	22.59 ± 0.12	23.56 ± 0.12	24.66 ± 0.12	24.54 ± 0.12	28.44	27.96	27.93	24.87 ± 0.12	32.24	33.91 ± 0.12	28.30	24.93 ± 0.12	17.96 ± 0.12	17.47 ± 0.12	20.58 ± 0.12
	Avg	20.13 ± 0.09	20.80 ± 0.09	20.49 ± 0.09	20.45 ± 0.09	24.52 ± 0.09	19.66 ± 0.09	20.58 ± 0.09	19.22 ± 0.09	19.77 ± 0.09	21.29 ± 0.09	19.80 ± 0.09	21.08 ± 0.09	19.58 ± 0.09	19.53 ± 0.09	20.54 ± 0.09	20.81 ± 0.09	19.63 ± 0.09	21.36 ± 0.09	22.89	22.54	22.92	21.11 ± 0.09	25.99	28.09 ± 0.12	23.64	21.32 ± 0.09	17.20 ± 0.09	16.67 ± 0.09	18.55 ± 0.09
MAPE (%)	3	30.74 ± 0.07	28.38 ± 0.07	29.73 ± 0.07	29.49 ± 0.07	55.76 ± 0.07	29.11 ± 0.07	30.04 ± 0.07	28.70 ± 0.07	30.52 ± 0.07	29.20 ± 0.07	29.69 ± 0.07	28.30 ± 0.07	29.13 ± 0.07	28.95 ± 0.07	31.70 ± 0.07	40.22 ± 0.07	29.02 ± 0.07	28.77 ± 0.07	28.04	27.81	28.80	27.03 ± 0.07	28.16	29.19 ± 0.07	28.44	27.36 ± 0.07	27.39 ± 0.07	26.07 ± 0.07	27.04 ± 0.07
	6	31.29 ± 0.08	29.84 ± 0.08	30.52 ± 0.08	30.31 ± 0.08	56.22 ± 0.08	30.00 ± 0.08	31.38 ± 0.08	29.47 ± 0.08	31.73 ± 0.08	30.11 ± 0.08	30.78 ± 0.08	28.55 ± 0.08	30.04 ± 0.08	29.84 ± 0.08	32.71 ± 0.08	41.21 ± 0.08	30.02 ± 0.08	30.57 ± 0.08	29.87	29.81	30.98	28.35 ± 0.08	30.54	31.69 ± 0.08	30.28	28.91 ± 0.08	27.74 ± 0.08	26.43 ± 0.08	27.75 ± 0.08
	12	33.45 ± 0.12	33.35 ± 0.12	32.93 ± 0.12	32.78 ± 0.12	52.24 ± 0.12	32.42 ± 0.12	34.68 ± 0.12	31.77 ± 0.12	35.13 ± 0.12	32.53 ± 0.12	30.28 ± 0.12	32.45 ± 0.12	32.43 ± 0.12	32.25 ± 0.12	35.68 ± 0.12	43.96 ± 0.12	32.40 ± 0.12	34.98 ± 0.12	35.06	35.14	35.86	31.18 ± 0.12	35.72	36.23 ± 0.12	34.51	32.29 ± 0.12	28.55 ± 0.12	27.19 ± 0.12	29.49 ± 0.12
	Avg	31.65 ± 0.08	30.22 ± 0.08	30.86 ± 0.08	30.65 ± 0.08	56.06 ± 0.08	30.30 ± 0.08	32.79 ± 0.08	29.80 ± 0.08	32.18 ± 0.08	30.41 ± 0.08	30.06 ± 0.08	29.52 ± 0.08	30.34 ± 0.08	30.14 ± 0.08	33.11 ± 0.08	41.57 ± 0.08	30.28 ± 0.08	31.07 ± 0.08	30.60	30.51	31.47	28.62 ± 0.08	30.97	32.02 ± 0.08	30.73	29.22 ± 0.08	27.83 ± 0.08	26.50 ± 0.08	27.95 ± 0.08

PEMS11

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	18.69 ± 0.1	17.88 ± 0.1	20.19 ± 0.1	18.90 ± 0.1	18.38 ± 0.1	17.03 ± 0.1	19.74 ± 0.1	16.62 ± 0.1	16.93 ± 0.1	22.48 ± 0.1	17.24 ± 0.1	23.71 ± 0.1	16.91 ± 0.1	16.76 ± 0.1	18.57 ± 0.1	20.18 ± 0.1	17.54 ± 0.1	18.32 ± 0.1	15.94	15.80	16.84	15.25 ± 0.0	18.69	20.45 ± 0.1	17.40	15.09 ± 0.1	16.33 ± 0.1	14.00 ± 0.1	14.52 ± 0.06
	6	19.19 ± 0.1	19.63 ± 0.1	20.93 ± 0.1	19.91 ± 0.1	18.10 ± 0.1	18.50 ± 0.1	21.22 ± 0.1	16.07 ± 0.1	18.43 ± 0.1	25.70 ± 0.1	18.77 ± 0.1	25.31 ± 0.1	18.36 ± 0.1	18.26 ± 0.1	19.61 ± 0.1	21.46 ± 0.1	18.98 ± 0.1	20.20 ± 0.1	19.31	19.21	20.39	17.88 ± 0.0	22.74	24.00 ± 0.1	20.82	17.34 ± 0.1	16.94 ± 0.1	14.90 ± 0.1	15.09 ± 0.06
	12	21.83 ± 0.1	22.48 ± 0.1	23.55 ± 0.1	22.77 ± 0.1	18.77 ± 0.1	21.61 ± 0.1	24.45 ± 0.1	20.92 ± 0.1	21.44 ± 0.1	28.19 ± 0.1	21.86 ± 0.1	29.22 ± 0.1	20.32 ± 0.1	21.38 ± 0.1	22.64 ± 0.1	24.40 ± 0.1	21.99 ± 0.1	24.43 ± 0.0	27.46	27.09	27.29	23.15 ± 0.0	29.56	30.42 ± 0.1	27.38	21.94 ± 0.1	18.28 ± 0.1	16.55 ± 0.1	19.05 ± 0.09
	Avg	19.69 ± 0.1	20.00 ± 0.1	21.35 ± 0.1	20.29 ± 0.1	18.54 ± 0.1	18.77 ± 0.1	21.54 ± 0.1	18.29 ± 0.1	18.67 ± 0.1	25.89 ± 0.1	19.02 ± 0.1	25.76 ± 0.1	18.60 ± 0.1	18.52 ± 0.1	20.02 ± 0.1	21.77 ± 0.1	19.24 ± 0.1	20.62 ± 0.1	20.23	20.04	20.91	18.32 ± 0.0	23.16	24.37 ± 0.0	21.24	17.71 ± 0.1	17.05 ± 0.1	14.99 ± 0.1	15.28 ± 0.07
RMSE	3	31.04 ± 0.1	29.94 ± 0.1	37.69 ± 0.1	31.36 ± 0.1	32.94 ± 0.1	28.37 ± 0.1	32.65 ± 0.1	27.81 ± 0.1	28.11 ± 0.1	42.15 ± 0.1	28.52 ± 0.1	37.06 ± 0.1	28.19 ± 0.1	27.97 ± 0.1	30.85 ± 0.1	32.56 ± 0.1	29.15 ± 0.1	30.60 ± 0.1	28.59	28.35	30.08	27.62 ± 0.0	33.70	36.68 ± 0.4	31.28	27.08 ± 0.1	27.05 ± 0.1	24.06 ± 0.1	25.11 ± 0.10
	6	32.77 ± 0.1	33.71 ± 0.1	39.44 ± 0.1	33.73 ± 0.1	36.32 ± 0.1	31.76 ± 0.1	35.91 ± 0.1	34.05 ± 0.1	31.50 ± 0.1	44.47 ± 0.1	31.98 ± 0.1	40.37 ± 0.1	31.47 ± 0.1	31.38 ± 0.1	33.20 ± 0.1	35.12 ± 0.1	32.37 ± 0.1	34.80 ± 0.1	35.64	35.45	37.19	33.61 ± 0.0	41.63	43.90 ± 0.2	37.82	32.06 ± 0.1	28.65 ± 0.1	26.27 ± 0.1	28.62 ± 0.12
	12	38.65 ± 0.1	41.33 ± 0.1	44.47 ± 0.1	39.49 ± 0.1	42.41 ± 0.1	38.27 ± 0.1	42.48 ± 0.1	36.98 ± 0.1	37.79 ± 0.1	49.10 ± 0.1	38.39 ± 0.1	47.69 ± 0.1	37.68 ± 0.1	37.94 ± 0.1	39.19 ± 0.1	40.62 ± 0.1	38.55 ± 0.1	43.44 ± 0.1	52.40	51.41	50.75	44.93 ± 0.1	54.25	56.11 ± 0.1	50.35	41.93 ± 0.1	31.70 ± 0.1	29.97 ± 0.1	34.75 ± 0.19
	Avg	33.66 ± 0.1	34.32 ± 0.1	40.11 ± 0.1	34.35 ± 0.1	36.95 ± 0.1	32.21 ± 0.1	36.45 ± 0.1	34.40 ± 0.1	31.90 ± 0.1	44.81 ± 0.1	32.38 ± 0.1	42.08 ± 0.1	31.88 ± 0.1	31.83 ± 0.1	33.89 ± 0.1	35.62 ± 0.1	32.79 ± 0.1	35.52 ± 0.1	37.49	37.05	38.14	34.43 ± 0.0	42.24	44.37 ± 0.1	38.56	32.78 ± 0.1	28.82 ± 0.1	26.40 ± 0.1	28.91 ± 0.13
MAPE (%)	3	30.78 ± 0.1	24.39 ± 0.1	29.45 ± 0.1	26.32 ± 0.1	27.32 ± 0.1	26.11 ± 0.1	28.92 ± 0.1	25.82 ± 0.1	26.53 ± 0.1	35.21 ± 0.1	28.98 ± 0.1	45.25 ± 0.1	27.10 ± 0.1	26.97 ± 0.1	30.64 ± 0.1	45.43 ± 0.1	29.58 ± 0.1	24.98 ± 0.1	21.78	21.66	22.80	20.81 ± 0.0	22.44	23.02 ± 0.0	22.43	20.95 ± 0.0	23.94 ± 0.1	21.39 ± 0.1	23.15 ± 0.27
	6	30.74 ± 0.1	26.12 ± 0.1	29.65 ± 0.1	27.14 ± 0.1	30.97 ± 0.1	27.25 ± 0.1	30.36 ± 0.1	25.01 ± 0.1	27.76 ± 0.1	36.53 ± 0.1	30.24 ± 0.1	46.71 ± 0.1	28.41 ± 0.1	27.11 ± 0.1	31.58 ± 0.1	47.41 ± 0.1	30.58 ± 0.1	26.94 ± 0.0	24.49	24.49	25.85	22.74 ± 0.0	25.41	26.42 ± 0.0	25.28	22.88 ± 0.0	24.36 ± 0.1	21.93 ± 0.1	24.21 ± 0.28
	12	32.84 ± 0.1	30.75 ± 0.1	32.76 ± 0.1	29.87 ± 0.1	31.84 ± 0.1	30.48 ± 0.1	34.14 ± 0.1	28.09 ± 0.1	31.40 ± 0.1	39.89 ± 0.1	33.90 ± 0.1	51.27 ± 0.1	31.92 ± 0.1	30.33 ± 0.1	35.10 ± 0.1	52.32 ± 0.1	33.38 ± 0.1	32.38 ± 0.5	31.67	31.70	32.22	26.79 ± 0.0	30.99	31.71 ± 0.0	31.06	26.96 ± 0.0	25.71 ± 0.1	23.19 ± 0.1	26.66 ± 0.30
	Avg	31.28 ± 0.1	26.69 ± 0.1	30.37 ± 0.1	27.55 ± 0.1	31.12 ± 0.1	27.67 ± 0.1	30.83 ± 0.1	25.39 ± 0.1	28.29 ± 0.1	36.90 ± 0.1	30.78 ± 0.1	47.34 ± 0.1	28.85 ± 0.1	27.53 ± 0.1	32.17 ± 0.1	48.00 ± 0.1	30.95 ± 0.1	27.66 ± 0.0	25.41	25.36	26.41	23.11 ± 0.0	25.78	26.67 ± 0.0	25.78	23.23 ± 0.0	24.54 ± 0.1	22.07 ± 0.1	24.48 ± 0.27

PEMS12

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-ZS	Moirai2-FM	STAEForm	STAE-Our	A2TTA
MAE	3	15.43 ± 0.02	14.23 ± 0.01	16.26 ± 0.03	16.16 ± 0.02	15.17 ± 0.01	13.59 ± 0.01	16.25 ± 0.02	13.24 ± 0.01	18.42 ± 0.04	14.97 ± 0.01	18.42 ± 0.01	13.96 ± 0.01	13.59 ± 0.01	13.41 ± 0.01	16.00 ± 0.01	17.97 ± 0.01	13.92 ± 0.01	14.31 ± 0.01	13.68	13.51	14.31	13.24 ± 0.01	15.91	17.32 ± 0.01	14.88	13.21 ± 0.01	12.39 ± 0.01	11.47 ± 0.01	12.34 ± 0.02
	6	15.76 ± 0.02	15.67 ± 0.01	16.80 ± 0.03	16.11 ± 0.02	16.28 ± 0.01	14.75 ± 0.01	17.78 ± 0.01	14.40 ± 0.01	14.60 ± 0.01	16.02 ± 0.01	14.57 ± 0.01	14.84 ± 0.01	14.72 ± 0.01	14.57 ± 0.01	16.76 ± 0.01	19.17 ± 0.01	15.06 ± 0.01	15.90 ± 0.01	16.23	16.05	17.06	15.37 ± 0.01	19.07	20.17 ± 0.01	17.51	15.06 ± 0.01	12.87 ± 0.01	12.04 ± 0.01	13.51 ± 0.03
	12	17.77 ± 0.03	13.87 ± 0.01	13.83 ± 0.01	16.46 ± 0.03	13.34 ± 0.01	17.02 ± 0.01	20.91 ± 0.01	16.58 ± 0.01	16.81 ± 0.01	13.11 ± 0.01	15.75 ± 0.01	16.63 ± 0.01	16.95 ± 0.01	16.87 ± 0.01	18.07 ± 0.02	22.03 ± 0.01	17.48 ± 0.01	19.37 ± 0.01	22.36	21.94	22.48	19.51 ± 0.01	24.73	25.32 ± 0.01	22.80	18.67 ± 0.01	16.70 ± 0.01	12.92 ± 0.01	15.63 ± 0.04
	Avg	16.16 ± 0.01	15.98 ± 0.01	17.13 ± 0.01	16.38 ± 0.01	16.41 ± 0.01	14.91 ± 0.01	18.04 ± 0.01	14.54 ± 0.01	14.74 ± 0.01	16.18 ± 0.01	14.71 ± 0.01	14.98 ± 0.01	14.88 ± 0.01	14.74 ± 0.01	16.07 ± 0.01	19.49 ± 0.01	15.27 ± 0.01	16.23 ± 0.01	16.92	16.67	17.48	15.69 ± 0.01	19.49	20.49 ± 0.01	17.90	15.32 ± 0.01	12.91 ± 0.01	12.05 ± 0.01	13.63 ± 0.03
RMSE	3	26.80 ± 0.03	25.07 ± 0.01	31.84 ± 0.05	26.13 ± 0.02	25.43 ± 0.01	23.87 ± 0.01	28.37 ± 0.01	23.41 ± 0.01	28.68 ± 0.01	26.16 ± 0.02	23.69 ± 0.01	24.14 ± 0.01	23.87 ± 0.01	23.58 ± 0.01	26.16 ± 0.01	29.50 ± 0.01	24.26 ± 0.01	25.36 ± 0.01	24.78	24.46	25.67	24.31 ± 0.01	29.08	32.10 ± 0.01	26.66	24.10 ± 0.01	22.03 ± 0.01	20.65 ± 0.01	22.16 ± 0.03
	6	27.76 ± 0.03	23.00 ± 0.01	32.92 ± 0.02	23.03 ± 0.01	27.77 ± 0.01	26.29 ± 0.01	30.54 ± 0.01	25.78 ± 0.01	26.12 ± 0.01	23.34 ± 0.01	26.08 ± 0.01	26.01 ± 0.01	26.23 ± 0.01	26.03 ± 0.01	27.64 ± 0.01	31.61 ± 0.01	26.59 ± 0.01	23.70 ± 0.01	29.93	29.57	31.07	29.01 ± 0.01	35.26	37.46 ± 0.01	31.59	28.07 ± 0.01	23.09 ± 0.01	21.87 ± 0.01	24.50 ± 0.04
	12	31.86 ± 0.03	24.05 ± 0.01	36.42 ± 0.02	32.38 ± 0.01	31.84 ± 0.01	30.73 ± 0.01	37.79 ± 0.01	29.93 ± 0.01	30.40 ± 0.01	32.39 ± 0.01	30.26 ± 0.01	29.33 ± 0.01	30.58 ± 0.01	30.50 ± 0.01	30.73 ± 0.01	35.73 ± 0.01	34.16 ± 0.01	35.51 ± 0.01	42.62	41.53	41.80	33.08 ± 0.01	46.15	47.62 ± 0.01	41.70	35.73 ± 0.01	24.77 ± 0.01	23.60 ± 0.01	23.45 ± 0.08
	Avg	28.46 ± 0.02	23.49 ± 0.01	33.43 ± 0.02	23.47 ± 0.01	27.96 ± 0.01	26.54 ± 0.01	32.01 ± 0.01	25.97 ± 0.01	26.32 ± 0.01	23.57 ± 0.01	26.28 ± 0.01	26.18 ± 0.01	26.48 ± 0.01	26.28 ± 0.01	28.14 ± 0.01	32.19 ± 0.01	26.92 ± 0.01	29.24 ± 0.01	31.42	30.85	31.92	29.74 ± 0.01	36.00	38.04 ± 0.01	32.32	28.62 ± 0.01	23.13 ± 0.01	21.85 ± 0.01	24.65 ± 0.04
MAPE (%)	3	30.24 ± 0.03	24.84 ± 0.01	29.71 ± 0.02	25.72 ± 0.01	27.23 ± 0.01	26.35 ± 0.01	27.74 ± 0.01	24.09 ± 0.01	25.52 ± 0.01	23.04 ± 0.01	24.99 ± 0.01	23.88 ± 0.01	25.59 ± 0.01	25.24 ± 0.01	24.60 ± 0.01	25.15 ± 0.01	27.92 ± 0.01	24.78 ± 0.01	22.56	22.43	23.78	21.67 ± 0.01	23.72	24.09 ± 0.01	23.33	21.79 ± 0.01	23.40 ± 0.01	21.53 ± 0.01	22.32 ± 0.33
	6	30.30 ± 0.03	26.69 ± 0.01	29.80 ± 0.02	26.82 ± 0.01	23.24 ± 0.01	26.56 ± 0.01	29.07 ± 0.01	25.13 ± 0.01	26.46 ± 0.01	29.04 ± 0.01	26.32 ± 0.01	29.31 ± 0.01	26.69 ± 0.01	26.45 ± 0.01	33.05 ± 0.01	28.10 ± 0.01	23.96 ± 0.01	27.01 ± 0.01	25.65	25.58	27.22	23.88 ± 0.01	27.23	28.45 ± 0.01	26.61	23.97 ± 0.01	23.81 ± 0.01	22.00 ± 0.01	23.52 ± 0.21
	12	32.77 ± 0.03	31.71 ± 0.01	36.40 ± 0.02	30.26 ± 0.01	30.60 ± 0.01	29.49 ± 0.01	32.30 ± 0.01	27.98 ± 0.01	29.56 ± 0.01	31.53 ± 0.01	29.50 ± 0.01	34.58 ± 0.01	29.76 ± 0.01	29.39 ± 0.01	37.84 ± 0.01	36.26 ± 0.01	34.77 ± 0.01	32.90 ± 0.01	34.04	33.94	34.72	28.54 ± 0.01	34.24	35.19 ± 0.01	33.62	28.62 ± 0.01	24.90 ± 0.01	23.08 ± 0.01	26.15 ± 0.07
	Avg	30.90 ± 0.02	27.32 ± 0.01	30.68 ± 0.02	27.32 ± 0.01	23.48 ± 0.01	26.88 ± 0.01	29.42 ± 0.01	25.48 ± 0.01	26.88 ± 0.01	29.30 ± 0.01	26.66 ± 0.01	29.71 ± 0.01	27.09 ± 0.01	26.77 ± 0.01	33.77 ± 0.01	28.93 ± 0.01	23.32 ± 0.01	27.74 ± 0.01	26.75	26.63	27.93	24.31 ± 0.01	27.76	28.81 ± 0.01	27.26	24.38 ± 0.01	23.94 ± 0.01	22.11 ± 0.01	23.78 ± 0.21

Headline takeaways (PEMS05)

Metric	Online-AN (prev best)	A2TTA-Lite (Ours)	Δ
Avg MAE	11.10	10.99	−0.11
Avg RMSE	17.63	17.53	−0.10
Avg MAPE	23.49	22.33	−1.16

Ablation (PEMS05 MAE, 5 seeds, mean $\pm$ std)

Variant	h3	h6	h12	Avg
backbone (Online-AN, frozen)	10.39 $\pm$ 0.07	11.00 $\pm$ 0.07	12.25 $\pm$ 0.11	11.10 $\pm$ 0.07
+ warmed-up calibrator (no online TTA)	10.52 $\pm$ 0.03	11.13 $\pm$ 0.04	12.35 $\pm$ 0.08	11.22 $\pm$ 0.04
TTA-Random	10.28 $\pm$ 0.05	10.89 $\pm$ 0.05	12.13 $\pm$ 0.09	10.99 $\pm$ 0.05
TTA-Recent	10.29 $\pm$ 0.05	10.91 $\pm$ 0.06	12.14 $\pm$ 0.09	11.00 $\pm$ 0.06
TTA-Error	10.28 $\pm$ 0.04	10.90 $\pm$ 0.04	12.14 $\pm$ 0.08	10.99 $\pm$ 0.04
A2TTA-Lite (full active)	10.29 $\pm$ 0.04	10.90 $\pm$ 0.03	12.14 $\pm$ 0.07	10.99 $\pm$ 0.04

TFNSW

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2	Chronos2	TimesFM	TimesFM-L	MoE-ZS	MoE-FM	Moirai2-Z	Moirai2-F	STAEForm	STAE-Our	A2TTA
MAE	3	168.71 ± 12.66	119.29 ± 14.12	127.73 ± 12.27	118.65 ± 11.48	114.38 ± 11.26	95.18 ± 10.08	123.61 ± 12.44	118.46 ± 11.29	119.39 ± 10.46	99.99 ± 13.28	127.68 ± 15.32	0.160.69	160.32	190.66	65.69 ± 0.255.63	88.08 ± 1.204.29	14.89 ± 0.88.89 ± 57.82 ± 0.44.11 ± 0.58												
	6	181.43 ± 14.51	111.00 ± 17.51	121.85 ± 14.23	118.04 ± 13.48	115.05 ± 12.73	112.15 ± 11.39	119.81 ± 11.39	119.74 ± 12.47	119.98 ± 15.11	115.13 ± 15.09	124.82 ± 19.15	1.244.74	245.53	282.31	134.82 ± 0.324.03	62.17 ± 0.301.81	15.95 ± 0.95.46 ± 76.00 ± 0.85.53 ± 0.59												
	12	191.66 ± 13.82	119.72 ± 20.35	126.70 ± 12.63	119.28 ± 14.80	116.43 ± 12.89	111.11 ± 16.71	115.43 ± 15.37	113.70 ± 16.07	116.78 ± 16.28	117.06 ± 16.10	126.10 ± 20.55	1.291.02	288.66	345.72	156.73 ± 0.359.28	71.62 ± 0.399.37	16.75 ± 0.107.50 ± 86.62 ± 0.11.92 ± 0.67												
	Avg	179.87 ± 14.01	118.33 ± 17.08	122.63 ± 16.08	114.36 ± 13.06	115.42 ± 12.41	117.86 ± 13.79	115.57 ± 14.44	114.41 ± 15.36	118.78 ± 14.40	118.22 ± 18.22	0.228.11	227.74	266.72	115.22 ± 0.306.03	34.59 ± 0.294.05	14.23 ± 0.96.12 ± 75.48 ± 0.65.10 ± 0.59													
RMSE	3	280.25 ± 20.76	171.81 ± 26.24	170.25 ± 24.50	169.38 ± 20.52	174.54 ± 22.92	172.97 ± 22.94	171.33 ± 21.16	172.93 ± 20.15	171.64 ± 22.26	177.14 ± 1.329.78	327.55	369.29	136.90 ± 0.462.48	61.94 ± 2.417.02	15.22 ± 0.171.62 ± 120.00 ± 0.67.36 ± 0.94														
	6	308.03 ± 28.64	194.80 ± 32.66	194.60 ± 22.75	193.40 ± 24.44	197.06 ± 27.26	196.33 ± 24.11	191.09 ± 25.08	193.32 ± 26.59	193.05 ± 28.06	196.95 ± 33.58	1.473.47	473.73	519.48	158.77 ± 1.570.16	49.78 ± 1.565.48	26.33 ± 0.189.36 ± 160.17 ± 0.96.80 ± 0.96													
	12	327.96 ± 28.20	197.82 ± 27.02	192.93 ± 28.37	196.29 ± 26.48	196.76 ± 34.68	197.74 ± 27.77	197.73 ± 27.92	190.96 ± 33.08	191.81 ± 29.09	195.84 ± 1.549.73	546.05	608.15	193.26 ± 0.648.38	52.88 ± 0.302.51	15.22 ± 0.209.12 ± 186.48 ± 0.26.88 ± 1.50														
	Avg	303.16 ± 29.72	190.75 ± 31.36	183.38 ± 22.75	181.17 ± 25.53	185.94 ± 34.86	185.92 ± 24.53	183.47 ± 26.79	180.719 ± 20.52	185.25 ± 25.72	181.13 ± 1.441.54	440.18	487.06	120.65 ± 0.544.56	109.59 ± 1.545.09	03.78 ± 0.190.33 ± 158.86 ± 0.91.81 ± 1.02														
MAPE (%)	3	203.98 ± 10.15	127.79 ± 10.98	119.02 ± 19.87	118.75 ± 12.81	117.14 ± 9.94	119.19 ± 7.34	118.83 ± 10.09	119.08 ± 12.93	119.03 ± 9.99	112.69 ± 142.98 ± 98.77 ± 554.16 ± 5.59.11	60.65	84.30	73.71 ± 1.5103.05	54.54 ± 0.868.13	43.39 ± 0.59.94 ± 56.28 ± 155.04 ± 2.36														
	6	210.50 ± 13.30	137.65 ± 13.75	127.56 ± 16.48	131.86 ± 10.02	130.84 ± 10.80	137.27 ± 12.40	135.70 ± 10.67	130.90 ± 12.83	131.49 ± 194.19 ± 6.116.26	117.43	163.00	128.42 ± 0.164.37	64.10 ± 1.131.57	55.49 ± 0.674.60 ± 61.23 ± 176.44 ± 2.48															
	12	221.79 ± 14.63	138.49 ± 14.63	138.74 ± 18.59	137.54 ± 16.14	147.11 ± 22.04	139.62 ± 18.22	132.49 ± 17.09	121.50 ± 21.14	120.61 ± 15.08	121.83 ± 4.180.70	182.97	242.48	171.51 ± 0.207.98	106.18 ± 1.226.64	10.90 ± 0.75.98 ± 70.66 ± 0.87.50 ± 2.52														
	Avg	211.72 ± 12.52	137.95 ± 13.84	130.83 ± 16.40	132.98 ± 9.93	136.92 ± 10.62	136.24 ± 14.80	131.93 ± 10.42	130.63 ± 13.22	132.00 ± 1613.97 ± 187.17 ± 5.119.00	120.67	160.45	124.25 ± 0.155.80	144.74 ± 1.137.97	62.69 ± 0.65.43 ± 62.06 ± 175.48 ± 2.43															

Ours improvement summary (new tta_ctx_local)

Dataset	MAE old→new (Δ , %)	RMSE old→new (Δ , %)	MAPE old→new (Δ , %)
PEMS03	13.25→12.59 (-0.66, -5.0%)	22.11→21.27 (-0.84, -3.8%)	24.65→23.41 (-1.24, -5.0%)
PEMS04	20.16→18.93 (-1.23, -6.1%)	31.82→30.60 (-1.22, -3.8%)	17.36→16.30 (-1.06, -6.1%)
PEMS05	10.82→10.48 (-0.34, -3.2%)	17.28→16.85 (-0.43, -2.5%)	22.70→21.49 (-1.21, -5.3%)
PEMS06	13.68→13.06 (-0.62, -4.5%)	21.30→20.44 (-0.86, -4.0%)	20.97→20.14 (-0.83, -4.0%)
PEMS07	16.42→15.65 (-0.77, -4.7%)	28.77→27.87 (-0.90, -3.1%)	24.30→21.98 (-2.32, -9.6%)
PEMS08	13.56→13.21 (-0.35, -2.6%)	22.77→22.34 (-0.43, -1.9%)	18.13→17.81 (-0.32, -1.8%)
PEMS10	11.40→11.06 (-0.34, -3.0%)	18.96→18.55 (-0.41, -2.2%)	28.68→27.95 (-0.73, -2.5%)
PEMS11	17.13→16.36 (-0.77, -4.5%)	29.97→28.84 (-1.13, -3.8%)	24.03→23.80 (-0.23, -1.0%)
PEMS12	14.15→13.63 (-0.52, -3.7%)	25.43→24.65 (-0.78, -3.1%)	24.04→23.78 (-0.26, -1.1%)
Mean	-4.1%	-3.1%	-4.0%

Ours improvement summary (new tta_ctx_local)

Dataset	OL-AN→Ours (MAE)	MAE↓	RMSE↓	MAPE↓
PEMS03	13.80→12.59	-8.8%	-7.1%	-6.2%
PEMS04	21.01→18.93	-9.9%	-6.5%	-8.6%
PEMS05	11.10→10.48	-5.6%	-4.4%	-8.5%
PEMS06	14.02→13.06	-6.8%	-5.7%	-8.2%
PEMS07	17.67→15.65	-11.4%	-7.5%	-17.6%
PEMS08	13.92→13.21	-5.1%	-3.7%	-7.6%
PEMS10	11.60→11.06	-4.7%	-3.5%	-6.2%
PEMS11	18.29→16.36	-10.6%	-8.1%	-6.3%
PEMS12	14.54→13.63	-6.3%	-5.1%	-6.7%
Mean	—	-7.7%	-5.7%	-8.4%